

Assignment 5A: Airline Delays

Mohd Afwan Shaikh

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Introduction and Approach

For this assignment, I analyzed airline arrival data for Alaska Airlines and AM West across five destinations: Los Angeles, Phoenix, San Diego, San Francisco, and Seattle.

I recreated the provided data in a CSV file using the original wide format and imported it into R. I checked the dataset for missing values and used dplyr and tidyr to transform it from wide format to long format.

I then calculated the percentage of flights that were delayed and on time for each airline. I compared the airlines' overall performance and examined their delay percentages separately across all five destinations using tables and visualizations.

Finally, I investigated the difference between the overall and city-by-city comparisons to understand how the distribution of flights across destinations affected the results.

Data Collection and Preparation

I recreated the airline arrival data from the assignment and saved it as a CSV file. The original dataset contains four rows representing the two arrival statuses for each airline and five columns containing destination-specific flight counts.

Data source: The airline arrival-delay table provided in the DATA 607 *Tidying and Transforming Data* assignment, originally attributed to *Numbersense* by Kaiser Fung (McGraw-Hill, 2013).

The dataset includes 20 flight counts covering two airlines, two arrival statuses, and five destinations. All 20 reported counts were present in the recreated CSV, so no missing values needed to be filled.

```
airline_data <- data.frame(
  Airline = c("Alaska", "Alaska", "AM West", "AM West"),
  Arrival = c("On Time", "Delayed", "On Time", "Delayed"),
  Los_Angeles = c(497, 62, 694, 117),
  Phoenix = c(221, 12, 4840, 415),
  San_Diego = c(212, 20, 383, 65),
  San_Francisco = c(503, 102, 320, 129),
  Seattle = c(1841, 305, 201, 61)
)

write.csv(
  airline_data,
  "airline_delays.csv",
  row.names = FALSE
)

print(airline_data)
```

```
##   Airline Arrival Los_Angeles Phoenix San_Diego San_Francisco Seattle
## 1  Alaska On Time      497      221      212          503      1841
## 2  Alaska Delayed       62       12       20          102      305
## 3 AM West On Time      694     4840      383          320      201
## 4 AM West Delayed     117      415       65          129       61
```

```
airline_data <- read.csv("airline_delays.csv")

head(airline_data)
```

```
##   Airline Arrival Los_Angeles Phoenix San_Diego San_Francisco Seattle
## 1  Alaska On Time      497      221      212          503      1841
## 2  Alaska Delayed       62       12       20          102      305
## 3 AM West On Time      694     4840      383          320      201
## 4 AM West Delayed     117      415       65          129       61
```

```
dim(airline_data)
```

```
## [1] 4 7
```

```
colSums(is.na(airline_data))
```

```
##      Airline      Arrival  Los_Angeles      Phoenix      San_Diego
##           0           0           0           0           0
## San_Francisco      Seattle
##           0           0
```

Checking for Missing Data

I checked the original dataset for missing values before transforming it. All reported flight counts were present, so no values needed to be replaced. If missing values are found in future data, I will investigate them before deciding how to handle them.

```
missing_before <- colSums(is.na(airline_data))

print(missing_before)
```

```
##      Airline      Arrival  Los_Angeles      Phoenix      San_Diego
##           0           0           0           0           0
## San_Francisco      Seattle
##           0           0
```

```
sum(is.na(airline_data))
```

```
## [1] 0
```

Demonstrating Missing-Value Handling

Although the recreated dataset contains no missing flight counts, the following example demonstrates how missing values could be populated when the source confirms that a missing count represents zero flights.

```
example_data <- airline_data

example_data$Phoenix[1] <- NA

example_data$Phoenix_Filled <- replace_na(
  example_data$Phoenix,
  0
)

print(
  example_data %>%
    select(Airline, Arrival, Phoenix, Phoenix_Filled)
)
```

```
##   Airline Arrival Phoenix Phoenix_Filled
## 1 Alaska On Time      NA              0
## 2 Alaska Delayed    12              12
## 3 AM West On Time  4840             4840
## 4 AM West Delayed  415              415
```

Data Validation

```
stopifnot(nrow(airline_data) == 4)
stopifnot(ncol(airline_data) == 7)
stopifnot(sum(is.na(airline_data)) == 0)
stopifnot(all(airline_data[, 3:7] >= 0))

cat("All data validation checks passed.")
```

```
## All data validation checks passed.
```

I validated the dataset by checking that it contained four rows and seven columns, had no missing values, and included no negative flight counts. All validation checks passed.

Transforming Data from Wide to Long

I used `pivot_longer()` to combine the five destination columns into a single Destination column. This produced a dataset with 20 rows and four columns, making it easier to compare flight counts across airlines and destinations.

```
airline_long <- airline_data %>%
  pivot_longer(
    cols = -c(Airline, Arrival),
    names_to = "Destination",
    values_to = "Flights"
  ) %>%
  mutate(
    Destination = gsub("_", " ", Destination)
  )

print(airline_long)
```

```
## # A tibble: 20 x 4
##   Airline Arrival Destination    Flights
##   <chr>    <chr>    <chr>      <int>
## 1 Alaska  On Time  Los Angeles    497
## 2 Alaska  On Time  Phoenix        221
## 3 Alaska  On Time  San Diego      212
## 4 Alaska  On Time  San Francisco  503
## 5 Alaska  On Time  Seattle       1841
## 6 Alaska  Delayed  Los Angeles     62
## 7 Alaska  Delayed  Phoenix         12
## 8 Alaska  Delayed  San Diego        20
## 9 Alaska  Delayed  San Francisco   102
## 10 Alaska Delayed  Seattle        305
## 11 AM West On Time  Los Angeles    694
## 12 AM West On Time  Phoenix      4840
## 13 AM West On Time  San Diego     383
## 14 AM West On Time  San Francisco  320
## 15 AM West On Time  Seattle       201
## 16 AM West Delayed  Los Angeles    117
```

```
## 17 AM West Delayed Phoenix      415
## 18 AM West Delayed San Diego    65
## 19 AM West Delayed San Francisco 129
## 20 AM West Delayed Seattle      61
```

Overall Airline Performance

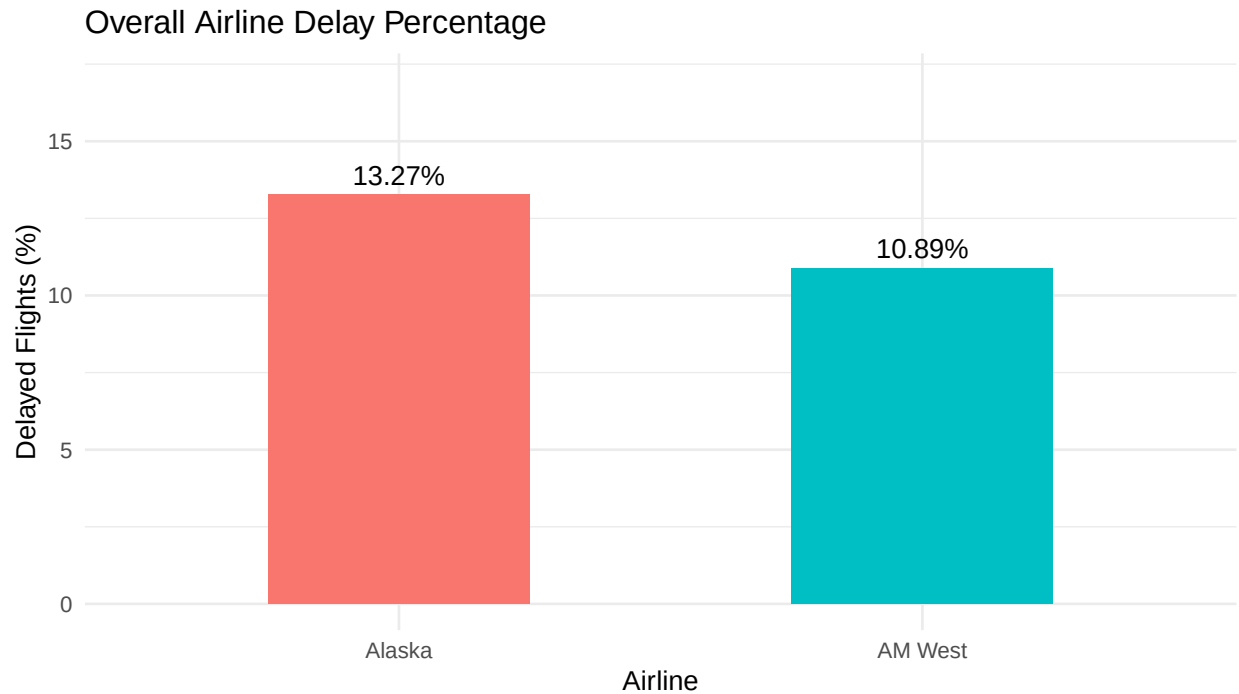
```
overall_performance <- airline_long %>%
  group_by(Airline, Arrival) %>%
  summarise(
    Flights = sum(Flights),
    .groups = "drop"
  ) %>%
  pivot_wider(
    names_from = Arrival,
    values_from = Flights
  ) %>%
  mutate(
    Total_Flights = `On Time` + Delayed,
    Delay_Percentage = round(Delayed / Total_Flights * 100, 2),
    On_Time_Percentage = round(`On Time` / Total_Flights * 100, 2)
  )

print(overall_performance)
```

```
## # A tibble: 2 x 6
##   Airline Delayed `On Time` Total_Flights Delay_Percentage On_Time_Percentage
##   <chr>      <int>    <int>      <int>          <dbl>          <dbl>
## 1 AM West    787      6438       7225          10.9           89.1
## 2 Alaska    501      3274       3775          13.3           86.7
```

Overall Delay Percentage

```
ggplot(
  overall_performance,
  aes(x = Airline, y = Delay_Percentage, fill = Airline)
) +
  geom_col(width = 0.5) +
  geom_text(
    aes(label = paste0(Delay_Percentage, "%")),
    vjust = -0.5
  ) +
  labs(
    title = "Overall Airline Delay Percentage",
    x = "Airline",
    y = "Delayed Flights (%)"
  ) +
  ylim(0, 17) +
  theme_minimal() +
  theme(legend.position = "none")
```



Airline Performance by Destination

```
city_performance <- airline_long %>%
  pivot_wider(
    names_from = Arrival,
    values_from = Flights
  ) %>%
  mutate(
    Total_Flights = `On Time` + Delayed,
    Delay_Percentage = round(Delayed / Total_Flights * 100, 2),
    On_Time_Percentage = round(`On Time` / Total_Flights * 100, 2)
  ) %>%
  arrange(Destination, Airline)

print(city_performance)
```

```
## # A tibble: 10 x 7
##   Airline Destination 'On Time' Delayed Total_Flights Delay_Percentage
##   <chr>      <chr>      <int>  <int>      <int>          <dbl>
## 1 AM West Los Angeles      694    117        811          14.4
## 2 Alaska  Los Angeles      497     62        559          11.1
## 3 AM West Phoenix      4840   415       5255           7.9
## 4 Alaska  Phoenix        221     12        233           5.15
## 5 AM West San Diego      383     65        448          14.5
## 6 Alaska  San Diego       212     20        232           8.62
## 7 AM West San Francisco   320    129        449          28.7
## 8 Alaska  San Francisco   503    102        605          16.9
## 9 AM West Seattle       201     61        262          23.3
```

```
## 10 Alaska Seattle 1841 305 2146 14.2
## # i 1 more variable: On_Time_Percentage <dbl>
```

Delay Percentage by Destination

```
ggplot(
  city_performance,
  aes(
    x = Destination,
    y = Delay_Percentage,
    fill = Airline
  )
) +
  geom_col(position = "dodge") +
  labs(
    title = "Airline Delay Percentage by Destination",
    x = "Destination",
    y = "Delayed Flights (%)",
    fill = "Airline"
  ) +
  theme_minimal()
```



Flight Distribution by Destination

The number of flights operated by each airline differs significantly across destinations. This helps explain why the overall delay percentages differ from the city-by-city results.

```
ggplot(
  city_performance,
  aes(x = Destination, y = Total_Flights, fill = Airline)
) +
  geom_col(position = "dodge") +
  labs(
    title = "Total Flights by Airline and Destination",
    x = "Destination",
    y = "Number of Flights",
    fill = "Airline"
  ) +
  theme_minimal()
```



Percentage of Flights by Destination

The following table shows the percentage of each airline's total flights operated to each destination.

```
flight_distribution <- city_performance %>%
  group_by(Airline) %>%
  mutate(
    Flight_Share = round(
      Total_Flights / sum(Total_Flights) * 100, 2
    )
  ) %>%
  ungroup()

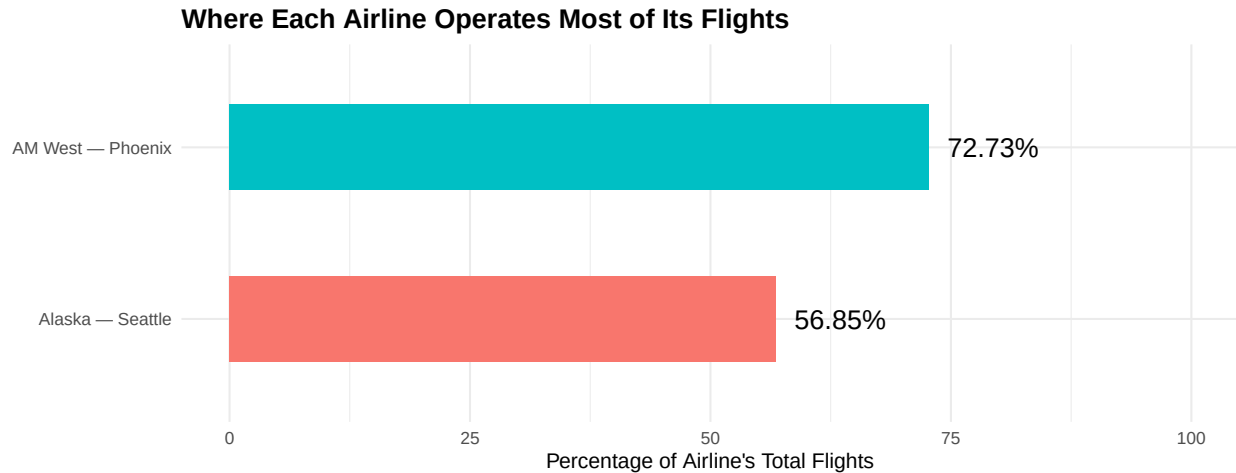
print(
  flight_distribution %>%
  select(Airline, Destination, Total_Flights, Flight_Share)
)
```

```
## # A tibble: 10 x 4
##   Airline Destination   Total_Flights Flight_Share
##   <chr>    <chr>         <int>      <dbl>
## 1 AM West Los Angeles      811        11.2
## 2 Alaska  Los Angeles      559        14.8
## 3 AM West Phoenix          5255       72.7
## 4 Alaska  Phoenix           233         6.17
## 5 AM West San Diego        448         6.2
## 6 Alaska  San Diego          232         6.15
## 7 AM West San Francisco    449         6.21
## 8 Alaska  San Francisco      605        16.0
## 9 AM West Seattle          262         3.63
## 10 Alaska Seattle         2146        56.8
```

Where Each Airline Operates Most of Its Flights

```
top_destinations <- flight_distribution %>%
  group_by(Airline) %>%
  slice_max(Flight_Share, n = 1, with_ties = FALSE) %>%
  ungroup() %>%
  mutate(
    Label = paste(Airline, Destination, sep = " - ")
  )

ggplot(
  top_destinations,
  aes(x = Flight_Share, y = Label, fill = Airline)
) +
  geom_col(width = 0.5) +
  geom_text(
    aes(label = paste0(Flight_Share, "%")),
    hjust = -0.2,
    size = 5
  ) +
  scale_x_continuous(limits = c(0, 100)) +
  labs(
    title = "Where Each Airline Operates Most of Its Flights",
    x = "Percentage of Airline's Total Flights",
    y = NULL
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", size = 14)
  )
```



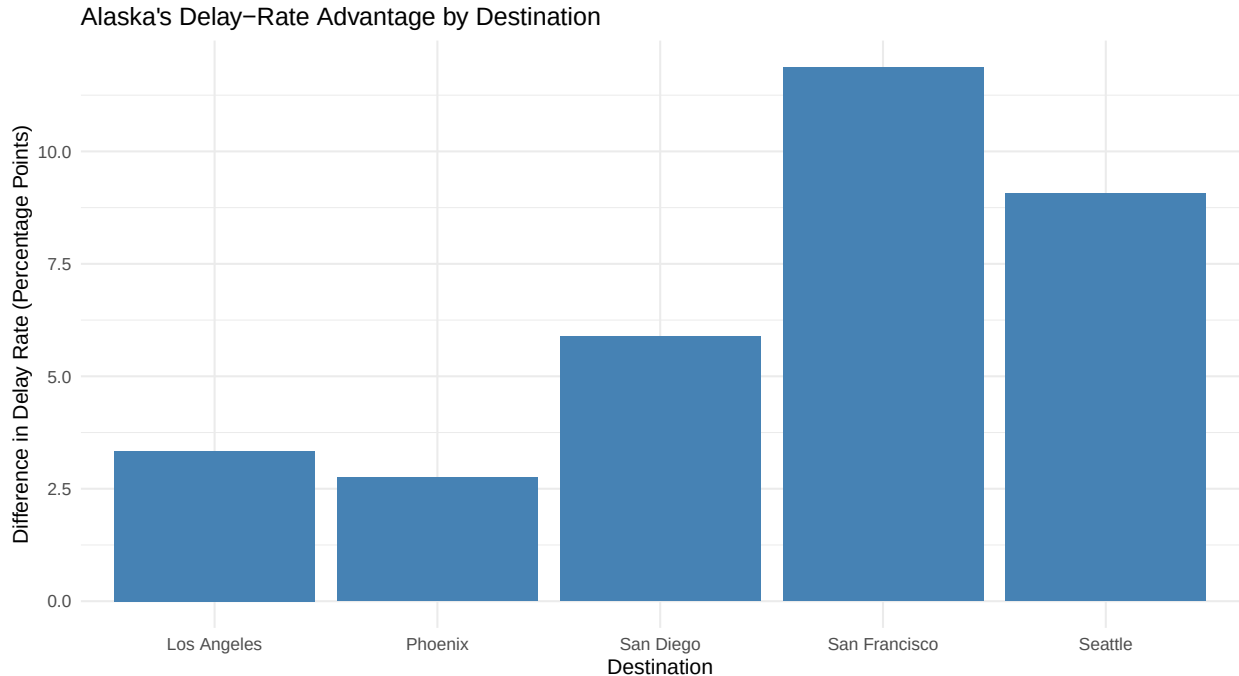
Difference in Delay Rates by Destination

```
delay_difference <- city_performance %>%
  select(Airline, Destination, Delay_Percentage) %>%
  pivot_wider(
    names_from = Airline,
    values_from = Delay_Percentage
  ) %>%
  mutate(
    Difference = `AM West` - Alaska
  )

print(delay_difference)
```

```
## # A tibble: 5 x 4
##   Destination 'AM West' Alaska Difference
##   <chr>         <dbl> <dbl>      <dbl>
## 1 Los Angeles    14.4  11.1       3.34
## 2 Phoenix         7.9   5.15       2.75
## 3 San Diego     14.5   8.62       5.89
## 4 San Francisco 28.7  16.9      11.9
## 5 Seattle       23.3  14.2       9.07
```

```
ggplot(
  delay_difference,
  aes(x = Destination, y = Difference)
) +
  geom_col(fill = "steelblue") +
  labs(
    title = "Alaska's Delay-Rate Advantage by Destination",
    x = "Destination",
    y = "Difference in Delay Rate (Percentage Points)"
  ) +
  theme_minimal()
```



Findings and Discussion

Overall Airline Performance

AM West had an overall delay rate of 10.89%, compared with 13.27% for Alaska Airlines. Based on the overall percentages alone, AM West appeared to perform better.

Performance Across the Five Destinations

When I compared the airlines separately across the five destinations, Alaska Airlines had a lower delay percentage in every city. The largest difference was in San Francisco, where Alaska had a delay rate of 16.86%, compared with 28.73% for AM West.

Explaining the Discrepancy

The difference between the overall and city-by-city comparisons is an example of Simpson's paradox. Although Alaska Airlines had a lower delay percentage in every destination, AM West had a lower overall delay percentage.

The flight distribution analysis helps explain this reversal. Approximately 72.73% of AM West's flights were to Phoenix, where both airlines had relatively low delay rates. In comparison, approximately 56.85% of Alaska's flights were to Seattle, where delays were more common.

Because the two airlines operated different proportions of flights across destinations, the overall comparison was influenced by their destination distributions. This demonstrates why examining percentages within individual groups is important when interpreting aggregated data.

Conclusion

This analysis showed that AM West had a lower overall delay percentage, while Alaska Airlines had a lower delay percentage in every individual destination. This reversal illustrates Simpson's paradox.

The results demonstrate that overall percentages can be misleading when groups have different distributions of observations. Comparing the airlines within each destination provided a more complete understanding of their performance.